1 離散数学・演習課題04 22059 鞠室

ex2.1

(1) プログラミング言語の名前 Scratch, Java, C, Python から成る集合

(2) 10より小さい非負整数の集合

(3) 単語 science に含まれる集合

(4) 自然数全体から成る集合

(5) $x^2=2$ を満たす整数の集合

ex2.2

例2.1(1)は外延的記法で{Scratch, Java, C, Python}と表される.

例2.1(4)は内包的記法で{x | xは自然数}

この集合をNとすれば {x | x∈N}と書ける.

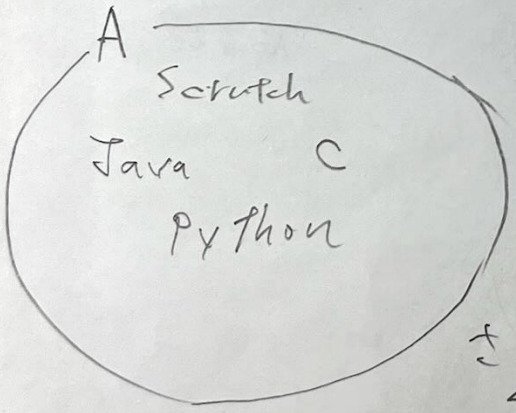
また「10以下の自然数から成る集合」を内包的記法で書くと

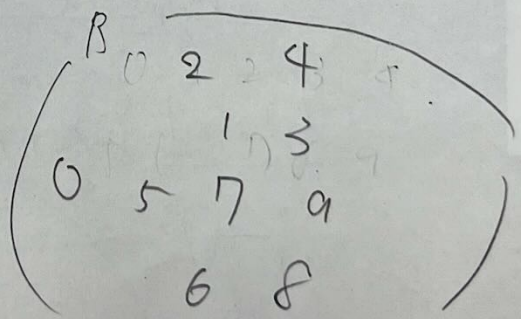
{x | x∈N かつ x≦10} あるいは {x∈N | x≦10}

ex2.3

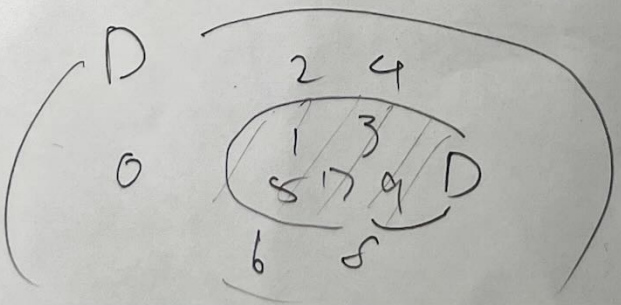
例2.1の(1)と(2)をそれぞれ集合A, Bとしてベン図を描く.

A={Scratch, Java, C, Python}　B={0, 1, 2, 3, 4, 5, 6, 7, 8, 9}





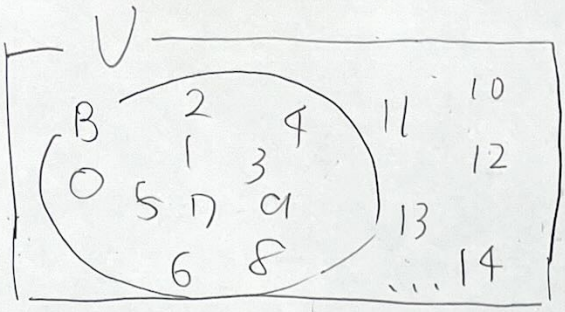
さらに, Bの要素の中で奇数の数だけから成る集合をDとしたときのベン図



ex 2.4

例2.1(2)を集合Bとすると全体集合Uが{x | xは非負整数}であるとき

このときのベン図は以下のようになる



ex 2.5

$|\{a, b, c, d\}| = 4$, $|\emptyset| = 0$, $|\{\{1, 2\}, \{1\}, \{2\}\}| = 3$

空集合$\emptyset$は要素が一つもないので濃度は0である

{1,2}と{1}と{2}が属しているので濃度は3である

Q 2.1

大きい石の集まり

Q 2.2

a) {0,1,2,3,4,5,6,7,8,9}

{s, c, i, e, n, ~~x~~, e}

b) (1) ~~集合をNとすると~~ {x | x∈N かつ x≦10}

xは整数

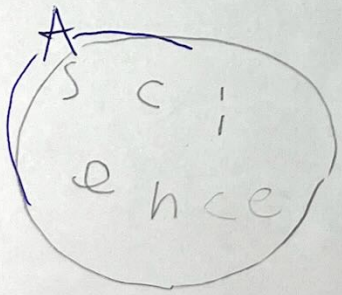
(2) ~~集合をNとすると~~ {x | x∈N かつ $x^2=2$}

xは整数

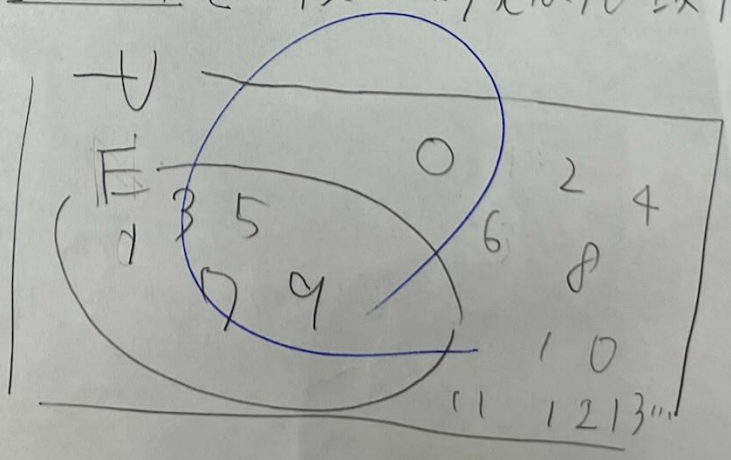
Q 2.3 {-2, -1, 0, 1, 2}

b, d e

Q 2.4



Q 2.5 {x | x∈N ; xは10以下の奇数}



Q2.6

有　無

(1)(2)(3)(5)　(2)(4)(5)

(1)(2)(3)(5)　(φ)

Q2.7

(1) 4　(2) 10　(3) 5　(5) 0

Q2.8

a) $|\{7,4,2,3\}| = 4$

b) $|\{\{1,2,3,4,5\}\}| = 1$

c) $|\{\}| = 0$

d) $|\{\phi, \{\phi\}\}| = 2$

Q2.9

「自然数の集合」は

{2,4,6}, {10,20}などであるが

「自然数全体の集合」は

Ｎただ1つだけ.

ex1.1

$1 \leqq U \leqq 15$

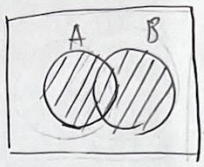
$x \div 3 = ? \cdots 1$

$X = \{1, 4, 7, 10, 13\}$

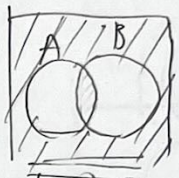
$\therefore X = \{x \in N : 1 \leqq x \leqq 15, 3 \mid (x-1)\}$

$X = \{3x+1 : x$は$0 \leqq x \leqq 4$となる整数$\}$

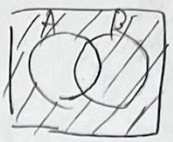
ex1.2　$\overline{A \cup B} = \bar{A} \cap \bar{B}$, $\overline{A \cap B} = \bar{A} \cup \bar{B}$



$A \cup B$

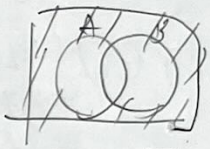


$\overline{A \cup B}$ =



$\bar{A}$

$\bar{B}$



$\bar{A} \cap \bar{B}$

$A \cap B$

$\overline{A \cap B}$ =

$\bar{A} \cup \bar{B}$

(1.1) $1 \leqq U \leqq 15$　2の倍数 ⇒ A

3の〃 ⇒ B

$C = \{x^2 - x + 1 : x$は$1 \leqq x \leqq 4$となる整数$\}$

(1) $A = \{2, 4, 6, 8, 10, 12, 14\}$

$B = \{3, 6, 9, 12, 15\}$

$C = \{1, 3, 7, 13\}$

(2) $A \setminus B = \{2, 4, 8, 10, 14\}$

$A \cap B = \{6, 12\}$, $\overline{A \cup B} = \{1, 5, 7, 11, 13\}$

(3) B, $A \cap B$, $\overline{A}$

$B = \{x \in U : 3 \mid x\}$

$A \cap B = \{x \in U : 6 \mid x\}$

$\overline{A} = \{1, 3, 5, 7, 9, 11, 13, 15\}$

$\therefore \overline{A} = \{x \in U : 2 \mid (x-1)\}$

2が $x-1$ を割り切れる

(4) $C \subseteq \overline{A \cup B}$

$\overline{A \cup B}$

1 5 7

11, 13

Cに3があるとの成り立たない

$3 \in C$, $3 \notin \overline{A \cup B}$

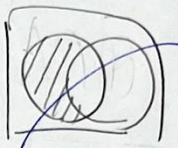
(2) $\{1, 4, 9, 16, 25\}$

(3) $x^2 + 3x = x(x+3)$

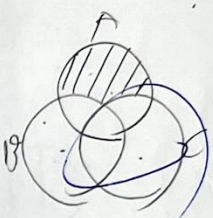
$\{0, 4, 10, 28, 88\}$

$\boxed{1.3}$

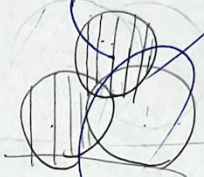
(1)



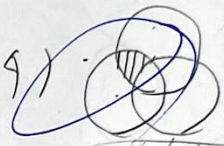
(2)



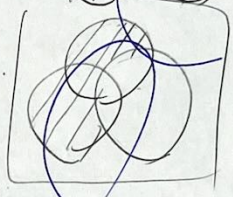
(3)



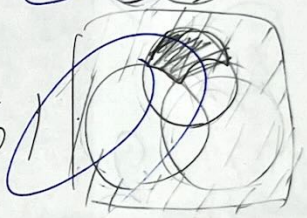
(4)



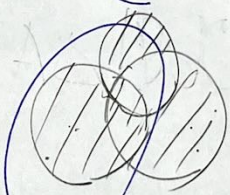
(5)



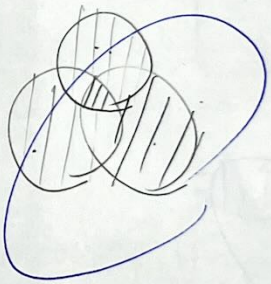
(6)



(7)



(8)



$\boxed{1.2}$

(1) $x^2 - 7x + 10 \leqq 0$

$(x-5)(x-2) \leqq 0$

$x = 2, 3, 4, 5$

$\therefore \{2, 3, 4, 5\}$